

Prevalence of physical activity and obesity in US counties, 2001–2011: a road map for action



Background Obesity and physical inactivity are associated with several chronic conditions, increased medical care costs, and premature death. Methods We used the Behavioral Risk Factor Surveillance System (BRFSS), a state-based random-digit telephone survey that covers the majority of United States counties, and the National Health and Nutrition Examination Survey (NHANES), a nationally representative sample of the US civilian noninstitutionalized population. About 3.7 million adults aged 20 years or older participated in the BRFSS from 2000 to 2011, and 30,000 adults aged 20 or older participated in NHANES from 1999 to 2010. We calculated body mass index (BMI) from self-reported weight and height in the BRFSS and adjusted for self-reporting bias using NHANES. We calculated self-reported physical activity—both any physical activity and physical activity meeting recommended levels—from self-reported data in the BRFSS. We used validated small area estimation methods to generate estimates of obesity and physical activity prevalence for each county annually for 2001 to 2011. Results Our results showed an increase in the prevalence of sufficient physical activity from 2001 to 2009. Levels were generally higher in men than in women, but increases were greater in women than men. Counties in Kentucky, Florida, Georgia, and California reported the largest gains. This increase in level of activity was matched by an increase in obesity in almost all counties during the same time period. There was a low correlation between level of physical activity and obesity in US counties. From 2001 to 2009, controlling for changes in poverty, unemployment, number of doctors per 100,000 population, percent rural, and baseline levels of obesity, for every 1 percentage point increase in physical activity prevalence, obesity prevalence was 0.11 percentage points lower. Conclusions Our study showed that increased physical activity alone has a small impact on obesity prevalence at the county level in the US. Indeed, the rise in physical activity levels will have a positive independent impact on the health of Americans as it will reduce the burden of cardiovascular diseases and diabetes. Other changes such as reduction in caloric intake are likely needed to curb the obesity epidemic and its burden.